

Kiran Kuruvila

kirankuruvila03@gmail.com | 73060 75537 | LinkedIn | GitHub | Portfolio

Professional Summary

Applied AI & Software Engineer combining strong mathematical and deep learning foundations with practical experience building backend APIs, real-time data streaming pipelines, and containerized cloud services. Proficient in Python, Java, Django, and FastAPI, with hands-on experience implementing neural architectures from first principles, building agentic LLM workflows (RAG, LangGraph), and deploying containerized services on AWS and Linux.

Education

Marian College, Kuttikkanam, Kerala

2025 – 2027

Master of Computer Applications (MCA)

MG University, Kottayam, Kerala

2022 – 2025

Bachelor of Arts in Economics

Technical Skills

Languages: Python, Java, SQL

Machine Learning & Deep Learning: PyTorch, Transformers, YOLOv8, CNNs, RNNs, GRUs/LSTMs, Vision Transformers (ViT), scikit-learn

LLM & Agentic AI: LangGraph, RAG, Semantic Vector Search, Embeddings, Function Calling, Structured State (Pydantic)

Backend & Streaming: FastAPI, Django, Django ORM, RESTful APIs, WebSockets, Server-Sent Events (SSE), Application Servers

Databases & Cloud: PostgreSQL, MySQL, SQL, Connection Pooling, AWS (EC2, ECR, RDS), Docker (Multi-stage builds), Linux, Nginx, Git, Postman

Mathematics & Engineering: Linear Algebra, Multivariate Calculus, Probability & Statistics, Stochastic Processes, OOP, Data Structures

Selected Projects

Real-Time Market Data & Quantitative ML Pipeline | *Python, WebSockets, FinBERT, NumPy, Pandas, REST APIs*

- Engineered an asynchronous live tick-ingestion pipeline via WebSockets, aggregating high-frequency trade feeds into 5-minute OHLC candles with boundary validation and sealed-state management.
- Built a fault-tolerant streaming architecture featuring heartbeat monitoring, tick-starvation anomaly detection, automated socket reconnection, and REST backfill for continuous data integrity.
- Computed real-time technical indicators (rolling RSI, EMA) in-memory alongside an applied FinBERT sentiment model to generate composite risk-on/risk-off momentum signals.

Recruitment Platform & AI Screening Service | *Django, PostgreSQL, Docker, AWS EC2, Nginx, RAG, Embeddings*

- Engineered a decoupled Django backend with PostgreSQL persistence and JWT authentication for candidate profiles, job postings, and application lifecycle tracking.
- Built an automated candidate screening pipeline converting unstructured PDF/DOCX resumes into schema-validated JSON, executing rubric match scoring in under 5 seconds end-to-end.
- Integrated a Recruiter Copilot powered by RAG and semantic vector search over candidate embeddings with calibrated token bounds (800–1500 tokens); containerized 3 microservices via multi-stage Docker builds on AWS EC2 behind an Nginx reverse proxy with SSL/TLS.

Multi-Agent Decision Backend & Visual Pattern Analysis | *FastAPI, LangGraph, Vision Models, Playwright, Docker, SSE*

- Architected a FastAPI backend orchestrating a 4-agent cyclical decision workflow (orchestration, risk assessment, and thesis re-evaluation) with Pydantic-validated state schemas and function calling.
- Built an automated headless chart extraction pipeline using Playwright with retry logic; served computer vision models to detect technical price action chart patterns from rendered charts.
- Implemented Server-Sent Events (SSE) to stream multi-agent reasoning traces to clients; containerized inference services with Docker for deployment.

From-Scratch Deep Learning & Neural Architectures | *Python, PyTorch, NumPy*

- Implemented foundational deep learning components from first principles: a scalar automatic differentiation engine (autograd) with reverse-mode backpropagation, alongside modular CNNs, RNNs, and LSTMs.
- Constructed a character-level NanoGPT and an arithmetic-prediction decoder Transformer from scratch, implementing multi-head causal self-attention, token embeddings, AdamW optimization, and learning-rate decay schedules.

Automated Market Gap Scanner | *Python, yfinance, Telegram Bot API, Cron, Linux*

- Developed an automated market gap scanner executing daily via Linux cron jobs, fetching live market data to filter stocks exceeding $\pm 1\%$ opening price thresholds.
- Dispatched structured gap analysis alerts to private Telegram channels using the Telegram Bot API with error handling and network retry mechanisms.

Teaching & Knowledge Sharing

Computer Science & Neural Foundations (Peer Instructor)

Marian College

- Conducted technical peer seminars teaching core computing foundations, object-oriented design, algorithm analysis, and gradient descent optimization from first principles.